Akshay Raj M

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Objective

M.Tech student in Data Science and AI with a strong interest in machine learning and data analysis. Seeking to apply programming and statistical methods to build practical ML solutions, analyze engineering data, and contribute to data-driven product decisions.

Experience

- **Keyvalue Software Systems, Ernakulam, Kerala** 15/07/2021 - 04/05/2024
  Test Engineer
  •Conducted functional, regression, sanity, and cross-browser testing for EdTech web and FinTech mobile applications, creating test plans, test cases, and defect reports using Jira; contributed to improving system test pass rates from 60% to 90% over two years.

  •Automated API validation workflows using Postman and Jest, improving test repeatability and reducing manual verification time.

  •Performed load testing with JMeter, analyzing latency/throughput/error trends to baseline performance and flag bottlenecks.

  •Used Git for version control and worked in AWS-based environments, collaborating with developers in iterative release cycles.

Education

- **Indian Institute of Technology, Tirupati** 2025-2027
  M.Tech Data Science and AI
  9.29
- **Govt Engineering College, Barton Hill, Trivandrum, Kerala** 2017-2021
  B.Tech Electronics and Communication
  9.22
- **Rahmaniya Higher Secondary School, Kozhikode, Kerala** 2015-2017
  Plus Two (HSE)
  97.9%
- **Technical Higher Secondary School, Vazhakkad, Malappuram, Kerala** 2015
  SSLC
  97.5%

Skills

- Programming: Python, C++
- ML / DL Frameworks: PyTorch, Scikit-learn, OpenCV
- Data Analysis: Pandas, NumPy, SQL, Exploratory Data Analysis
- Visualization: Matplotlib, Seaborn, Power BI
- Concepts: Machine Learning, Deep Learning, GANs, Large Language Models (LLMs).
- Tools: Git, Linux, AWS

Projects

- **Deep Reinforcement Learning for Humanoid Locomotion**
  To develop and end-to-end pipeline that enables a simulated humanoid agent to learn stable walking using Deep Reinforcement Learning
- **Geospatial deep learning for land classification**
  Geospatial Deep Learning for Land Classification
  Implemented a ResNet-18 transfer learning model in PyTorch to classify satellite imagery using the ESA WorldCover dataset, achieving ~93% accuracy.
- **Image Denoising using NAFNet**
  Implemented the denoising component of NAFNet(Non linear Activation free network) proposed in ECCV 2022 for image restoration tasks. The model was fine tuned and evaluated the performance.
- **Human Detection and Multi-Object Tracking in Video**
  Developed a computer vision pipeline to detect and track humans in video streams using YOLOv8 object detection model.

Relevant Coursework

- Machine learning | Deep learning | Data Science and Engineering | Linear Algebra and Probability | Multivariate Statistics